NEET UG 2027: Physics Formula Compendium

Module 02: Rotational Dynamics, Gravitation, Solids, Fluids, Thermal & Oscillations

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Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Rotational Dynamics & Centre of Mass (Unit 5)

1.1 Centre of Mass (CM) Formulations

Discrete and Continuous Centre of Mass Coordinates

$$\mathbf{r}_{\text{cm}} = \frac{\sum_{i=1}^{n} m_i \mathbf{r}_i}{\sum_{i=1}^{n} m_i} = \frac{1}{M}\sum m_i \mathbf{r}_i, \qquad \mathbf{r}_{\text{cm}} = \frac{\int \mathbf{r}\, dm}{\int dm} = \frac{1}{M}\int \mathbf{r}\, dm \tag{1}$$

Centre of Mass Velocity, Acceleration & Truncated Body Relation:

$$\mathbf{v}_{\text{cm}} = \frac{\sum m_i \mathbf{v}_i}{M}, \quad \mathbf{a}_{\text{cm}} = \frac{\mathbf{F}_{\text{ext}}}{M}, \quad \mathbf{r}_{\text{rem}} = \frac{M\mathbf{r}_{\text{orig}} - m\mathbf{r}_{\text{cut}}}{M - m} \tag{2}$$

Geometry / Symmetric Body	CM Location (y_{cm})	Geometry / Symmetric Body	CM Location (y_{cm})
Uniform Semicircular Ring	$\frac{2R}{\pi}$ from base	Uniform Semicircular Disc	$\frac{4R}{3\pi}$ from base
Hemispherical Shell (Hollow)	$\frac{R}{2}$ from center	Solid Hemisphere	$\frac{3R}{8}$ from center
Hollow Cone (without base)	$\frac{h}{3}$ from base	Solid Cone	$\frac{h}{4}$ from base

Table 1: Exact Centre of Mass Coordinates from Base / Symmetric Center along Axis.

1.2 Rotational Kinematics, Dynamics & Angular Momentum

Governing Torque and Angular Momentum Relations

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = I\boldsymbol{\alpha} = \frac{d\mathbf{L}}{dt} \qquad \text{[Torque Equation; } \tau = rF\sin\theta\text{]} \tag{3}$$

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = I\boldsymbol{\omega} \qquad \text{[Angular Momentum of Rigid Body]} \tag{4}$$

$$W_{\text{rot}} = \int \tau\, d\theta = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 \qquad \text{[Rotational Work-Energy Theorem]} \tag{5}$$

Conservation of Angular Momentum (When $\boldsymbol{\tau}_{\text{ext}} = 0$):

$$\mathbf{L} = I_1\boldsymbol{\omega}_1 = I_2\boldsymbol{\omega}_2 = \text{constant} \implies \omega \propto \frac{1}{I} \tag{6}$$

1.3 Moment of Inertia Theorems & Standard Geometrical Values

Moment of Inertia (MI) Fundamental Theorems

$$I = \sum m_i r_i^2 = \int r^2\, dm = Mk^2 \quad [k = \text{Radius of Gyration}] \tag{7}$$

$$I = I_{\text{cm}} + Md^2 \quad [\text{Parallel Axis Theorem: valid for ANY rigid body}] \tag{8}$$

$$I_z = I_x + I_y \quad [\text{Perpendicular Axis Theorem: strictly PLANAR laminar bodies only}] \tag{9}$$

Body (Mass M, Radius R)	Axis of Rotation	Moment of Inertia (I)	k^2/R^2
Thin Rod (Length L)	Perpendicular through center	$\frac{1}{12}ML^2$	$L^2/(12R^2)$
Thin Rod (Length L)	Perpendicular through one end	$\frac{1}{3}ML^2$	$L^2/(3R^2)$
Thin Circular Ring	Central perpendicular axis	MR^2	1
Thin Circular Ring	Across diameter	$\frac{1}{2}MR^2$	1/2
Uniform Circular Disc	Central perpendicular axis	$\frac{1}{2}MR^2$	1/2
Uniform Circular Disc	Across diameter	$\frac{1}{4}MR^2$	1/4
Solid Cylinder	Longitudinal central axis	$\frac{1}{2}MR^2$	1/2
Hollow Cylinder (Thin Pipe)	Longitudinal central axis	MR^2	1
Solid Sphere	Through any diameter	$\frac{2}{5}MR^2$	$2/5 = 0.40$
Spherical Shell (Hollow Sphere)	Through any diameter	$\frac{2}{3}MR^2$	$2/3 \approx 0.67$

Table 2: Standard Moments of Inertia and Dimensionless Gyration Factor k^2/R^2.

1.4 Pure Rolling Dynamics (Rolling Without Slipping)

Rolling Motion Governing Equations

$$v_{\text{cm}} = R\omega, \qquad a_{\text{cm}} = R\alpha \quad [\text{No-slip Kinematic Constraint}] \tag{10}$$

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2\left(1 + \frac{k^2}{R^2}\right) \tag{11}$$

$$\frac{K_{\text{trans}}}{K_{\text{total}}} = \frac{1}{1 + k^2/R^2}, \qquad \frac{K_{\text{rot}}}{K_{\text{total}}} = \frac{k^2/R^2}{1 + k^2/R^2} \tag{12}$$

Rolling Down an Incline of Angle θ and Height h:

$$a = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}}, \qquad v_{\text{bottom}} = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}, \qquad \mu_{\min} = \frac{\tan\theta}{1 + \frac{R^2}{k^2}} \tag{13}$$

Race Down an Incline (Identical M, R, θ, h)

- Ordering of k^2/R^2: Solid Sphere (0.40) < Disc/Cylinder (0.50) < Hollow Sphere (0.67) < Ring/Hollow Cylinder (1.00).
- Acceleration Order: $a_{\text{solid sphere}} > a_{\text{disc}} > a_{\text{hollow sphere}} > a_{\text{ring}}$.
- Time of Descent ($t = \sqrt{2s/a}$): $t_{\text{solid sphere}} < t_{\text{disc}} < t_{\text{hollow sphere}} < t_{\text{ring}}$. Solid sphere reaches first!
- Velocity at Bottom: $v_{\text{solid sphere}} > v_{\text{disc}} > v_{\text{hollow sphere}} > v_{\text{ring}}$.

Rotational Traps: Friction & Perpendicular Axis Theorem

- **Friction in Pure Rolling:** Static friction acts at the instantaneous point of contact. Because the point of contact is instantaneously at rest ($v_{\text{contact}} = 0$), **work done by friction in pure rolling is identically zero** ($W_f = 0$). Mechanical energy is conserved.
- **Perpendicular Axis Theorem Trap:** Cannot be applied to 3D bodies like spheres, solid cylinders, or cones. It is strictly valid for 2D plane laminar sheets ($I_z = I_x + I_y$).

2 Gravitation (Unit 6)

2.1 Universal Gravitation, Acceleration (g) & Field Variations

Newton's Law & Surface Acceleration g

$$F = \frac{GM_1M_2}{r^2} \qquad [G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}] \tag{14}$$

$$g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R \qquad [\text{Surface acceleration, Earth } g \approx 9.8\,\mathrm{m\,s^{-2}}] \tag{15}$$

Variations of g (Altitude, Depth & Latitude):

$$g_h = g\left(\frac{R}{R+h}\right)^2 = g\left(1+\frac{h}{R}\right)^{-2} \approx g\left(1-\frac{2h}{R}\right) \qquad [\text{Approximation valid strictly for } h \ll R] \tag{16}$$

$$g_d = g\left(1-\frac{d}{R}\right) \qquad [\text{Exact at all depths } 0 \le d \le R] \tag{17}$$

$$g' = g - \omega^2 R\cos^2\lambda \qquad [\lambda = \text{latitude; Equator } \lambda = 0, \text{Pole } \lambda = 90^\circ] \tag{18}$$

Exact Comparison & Extreme Earth Rotation Cases

- **Equal Reduction at Altitude and Depth:** For small $h \ll R$, $g_h = g_d \implies d = 2h$. A height of 1km causes the same reduction in g as a depth of 2km.
- **Weightlessness at Equator:** For bodies at equator to become weightless ($g' = 0$):

$$\omega = \sqrt{\frac{g}{R}} \approx \frac{1}{800}\mathrm{rad\,s^{-1}}, \qquad T = 2\pi\sqrt{\frac{R}{g}} \approx 84.6 \text{ minutes} \approx 1.41 \text{ hours}$$

  This requires the Earth to rotate roughly **17 times faster** than its current angular velocity.
- **Earth Stops Rotating:** If $\omega \to 0$, g increases by $\omega^2 R$ at the equator, but **remains unchanged at the poles**.

2.2 Gravitational Potential, Escape & Orbital Velocities

Potential, Escape Speed, Orbital Speed & Satellite Energies

$$V(r) = -\frac{GM}{r}, \qquad U(r) = -\frac{GMm}{r} \qquad [E_g = -\frac{dV}{dr} = -\frac{GM}{r^2}] \tag{19}$$

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2\,\text{km}\,\text{s}^{-1} \qquad \text{[Escape Velocity from Earth Surface]} \tag{20}$$

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R+h}} \qquad \text{[Orbital Velocity at orbit radius } r = R+h\text{]} \tag{21}$$

$$v_e = \sqrt{2}v_o \approx 1.414 v_o \qquad \text{[Near-Earth orbit } h \ll R : v_o \approx 7.92\,\text{km}\,\text{s}^{-1}\text{]} \tag{22}$$

$$T^2 = \left(\frac{4\pi^2}{GM}\right) r^3 \implies T = 2\pi\sqrt{\frac{r^3}{GM}} \qquad \text{[Kepler's Third Law of Periods]} \tag{23}$$

Satellite Energy Spectrum (Orbit radius $r = R + h$):

$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}, \quad E_{\text{total}} = -\frac{GMm}{2r} = -K = \frac{U}{2}, \quad \text{B.E.} = -E_{\text{total}} = \frac{GMm}{2r} \tag{24}$$

Gravitation Traps: Launch Angle & Solid Sphere Potential

- **Escape Velocity Independence:** Escape velocity $v_e = \sqrt{2gR}$ is **completely independent** of the mass of the projectile m and the angle of launch θ. A body launched at $30°, 60°$ or $90°$ requires the identical escape speed $11.2\,\text{km}\,\text{s}^{-1}$.
- **Gravitational Potential Inside Uniform Solid Sphere (Earth):**

$$V_{\text{in}}(r) = -\frac{GM}{2R^3}\left(3R^2 - r^2\right) \implies V_{\text{center}} = -\frac{3}{2}\frac{GM}{R} = 1.5\,V_{\text{surface}}$$

  Potential is NOT zero at the centre; its magnitude is 1.5 times the surface value.

3 Properties of Solids & Fluids (Unit 7)

3.1 Elasticity & Moduli of Elasticity

Hooke's Law, Elastic Moduli & Energy Density

$$Y = \frac{\text{Longitudinal Stress}}{\text{Longitudinal Strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L} \quad [\text{Young's Modulus, } \Delta L = \frac{FL}{AY}] \tag{25}$$

$$B = -\frac{\Delta P}{\Delta V/V} = -V\frac{\Delta P}{\Delta V}, \qquad K = \frac{1}{B} \quad [\text{Bulk Modulus } B \text{ \& Compressibility } K] \tag{26}$$

$$\eta = \frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{F/A}{\theta} \quad [\text{Shear Modulus / Modulus of Rigidity}] \tag{27}$$

$$\sigma = -\frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = -\frac{\Delta d/d}{\Delta L/L} \quad [\text{Poisson's Ratio: Theoretical: } -1 \le \sigma \le 0.5\text{; Actual: } 0 < \sigma < 0.5] \tag{28}$$

Elastic Potential Energy Stored in Stretched Wire:

$$U = \frac{1}{2}F\Delta L = \frac{1}{2}k(\Delta L)^2, \qquad u = \frac{U}{V} = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{(\text{stress})^2}{2Y} = \frac{1}{2}Y(\text{strain})^2 \tag{29}$$

Thermal Stress Developed in Clamped Rod:

$$\text{Stress} = Y\alpha\Delta T, \qquad F_{\text{tension}} = YA\alpha\Delta T \tag{30}$$

3.2 Fluid Statics, Pascal's Principle & Buoyancy

Pressure, Pascal's Law & Archimedes' Principle

$$P = P_0 + \rho g h \quad [\text{Gauge Pressure } P_{\text{gauge}} = \rho g h] \tag{31}$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \implies F_2 = F_1\left(\frac{A_2}{A_1}\right) \quad [\text{Hydraulic Lift Principle}] \tag{32}$$

$$F_B = V_{\text{submerged}}\,\rho_{\text{liquid}}\,g \quad [\text{Archimedes' Buoyant Force}] \tag{33}$$

$$\frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{body}}}{\rho_{\text{liquid}}}, \qquad W_{\text{apparent}} = W_{\text{actual}}\left(1 - \frac{\rho_{\text{liquid}}}{\rho_{\text{body}}}\right) \tag{34}$$

3.3 Fluid Dynamics, Viscosity & Surface Tension

Continuity, Bernoulli, Viscosity & Capillarity

$$A_1 v_1 = A_2 v_2 = \text{constant} \quad \text{[Equation of Continuity for Incompressible Fluid]} \tag{35}$$

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant} \quad \text{[Bernoulli's Theorem]} \tag{36}$$

$$v_{\text{efflux}} = \sqrt{2gh}, \quad R_{\text{horizontal}} = 2\sqrt{h(H-h)} \quad \text{[Torricelli's Law; } R_{\max} = H \text{ at } h = H/2\text{]} \tag{37}$$

$$F_v = 6\pi\eta r v \quad \text{[Stokes' Viscous Drag Law]} \tag{38}$$

$$v_t = \frac{2}{9}\frac{r^2(\rho-\sigma)g}{\eta} \implies v_t \propto r^2 \quad \text{[Terminal Velocity; } \rho = \text{body density}, \sigma = \text{fluid density}\text{]} \tag{39}$$

$$R_e = \frac{\rho v d}{\eta} \quad \text{[Reynolds Number: } R_e < 2000 \text{ Laminar; } R_e > 3000 \text{ Tur} \tag{40}$$

Surface Tension, Excess Pressure & Capillary Rise:

$$T = \frac{F}{L} = \frac{\Delta W}{\Delta A}, \qquad W_{\text{splitting}} = 4\pi R^2 T(n^{1/3} - 1) \quad \text{[1 big drop} \to n \text{ identical droplets]} \tag{41}$$

$$\Delta P_{\text{liquid drop}} = \frac{2T}{R}, \qquad \Delta P_{\text{soap bubble}} = \frac{4T}{R} \quad \text{[Soap bubble has TWO free air-liquid interfaces]} \tag{42}$$

$$h = \frac{2T\cos\theta}{r\rho g} = \frac{2T}{R_c \rho g} \quad \text{[Jurin's Law: } hr = \text{constant; } R_c = r/\cos\theta \text{ is meni} \tag{43}$$

Fluid Traps: Terminal Velocity & Insufficient Capillary Length

- **Terminal Velocity (r^2 vs r):** Viscous force $F_v \propto r$, but terminal velocity $v_t \propto r^2$. If two drops of radii $1:2$ coalesce, the radius becomes $R = 2^{1/3}r$; new terminal velocity increases by a factor of $2^{2/3}$.
- **Insufficient Capillary Tube Length ($l < h$):** Liquid **never overflows** from a capillary tube of insufficient height l. Instead, the liquid reaches the upper edge and increases its meniscus radius of curvature R_c' such that $hr = lR_c'$ ($R_c' > R_c$).

4 Thermal Physics & Thermodynamics (Units 7, 8 & 9)

4.1 Thermal Expansion, Calorimetry & Heat Transfer

Expansion, Calorimetry, Conduction & Radiation

$$\Delta L = L_0\alpha\Delta T, \quad \Delta A = A_0\beta\Delta T, \quad \Delta V = V_0\gamma\Delta T \quad [\alpha:\beta:\gamma = 1:2:3] \tag{44}$$

$$Q = mc\Delta T, \qquad Q = mL \quad [c = \text{sp. heat capacity}, L = \text{latent heat}] \tag{45}$$

$$\frac{dQ}{dt} = \frac{kA(T_1 - T_2)}{L} = \frac{\Delta T}{R_{\text{th}}} \quad [R_{\text{th}} = \frac{L}{kA} = \text{Thermal Resistance}] \tag{46}$$

$$E = \sigma e A T^4, \qquad P_{\text{net}} = \sigma e A(T^4 - T_0^4) \quad [\text{Stefan-Boltzmann Law}, \sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}] \tag{47}$$

$$\lambda_m T = b = 2.898 \times 10^{-3}\,\text{m K} \quad [\text{Wien's Displacement Law}] \tag{48}$$

$$\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_0\right) \quad [\text{Newton's Law of Cooling (Average Form)}] \tag{49}$$

4.2 First Law of Thermodynamics & Process Formulations

First Law & Thermodynamic Processes

$$\Delta Q = \Delta U + \Delta W \quad [\text{Ideal Gas: } \Delta U = nC_v\Delta T = \frac{P_2V_2 - P_1V_1}{\gamma - 1} \text{ for ANY process}] \tag{50}$$

Work Done in Standard Thermodynamic Processes:

$$W_{\text{isochoric}} = 0, \qquad \Delta Q = \Delta U = nC_v\Delta T \quad [V = \text{constant}] \tag{51}$$

$$W_{\text{isobaric}} = P(V_2 - V_1) = nR(T_2 - T_1), \quad \Delta Q = nC_p\Delta T \quad [P = \text{constant}] \tag{52}$$

$$W_{\text{isothermal}} = nRT\ln\left(\frac{V_2}{V_1}\right) = 2.303nRT\log_{10}\left(\frac{P_1}{P_2}\right) \quad [T = \text{const}, \Delta U = 0, \Delta Q = W] \tag{53}$$

$$W_{\text{adiabatic}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U \quad [\Delta Q = 0, PV^\gamma = \text{const}] \tag{54}$$

Gas Atomicity	Deg. of Freedom (f)	$C_v = \frac{f}{2}R$	$C_p = C_v + R$	$\gamma = C_p/C_v = 1 + \frac{2}{f}$
Monoatomic (He, Ne, Ar)	3 (translational)	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$
Diatomic (Rigid: O_2, N_2)	3 trans + 2 rot = 5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.40$
Non-Linear Polyatomic (H_2O, NH_3)	3 trans + 3 rot = 6	$3R$	$4R$	$\frac{4}{3} \approx 1.33$

Table 3: Molar Heat Capacities and Adiabatic Exponents for Standard Gases.

High-Yield Thermodynamic Relations & Polytropic Processes

- **Energy Partition in Isobaric Process:**

$$\Delta W : \Delta U : \Delta Q = nR\Delta T : nC_v\Delta T : nC_p\Delta T = (\gamma - 1) : 1 : \gamma$$

For a diatomic gas ($\gamma = 7/5$), fraction of heat converted to work is $\frac{\Delta W}{\Delta Q} = \frac{\gamma - 1}{\gamma} = \frac{2}{7} \approx 28.6\%$.

- **General Polytropic Process ($PV^n = \text{const}$):**

$$W = \frac{nR(T_1 - T_2)}{n-1}, \qquad C = C_v + \frac{R}{1-n}$$

- **Comparison of $P-V$ Curves:** $\left(\frac{dP}{dV}\right)_{\text{adiabatic}} = \gamma \left(\frac{dP}{dV}\right)_{\text{isothermal}}$. Adiabatic slope is γ-times steeper!

4.3 Kinetic Theory of Gases (KTG)

Pressure, Speeds & Mean Free Path

$$P = \frac{1}{3}\rho v_{\text{rms}}^2 = \frac{1}{3}\frac{M}{V}v_{\text{rms}}^2, \qquad E_{\text{total, translation}} = \frac{3}{2}nRT = \frac{3}{2}Nk_BT \tag{55}$$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}, \quad v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}, \quad v_{\text{mp}} = \sqrt{\frac{2RT}{M}} \implies v_{\text{mp}} : v_{\text{avg}} : v_{\text{rms}} = 1 : 1.128 : 1.224 \tag{56}$$

$$\lambda = \frac{1}{\sqrt{2}\pi n d^2} = \frac{k_BT}{\sqrt{2}\pi d^2 P} \quad \left[\lambda \propto \frac{T}{P}; \text{if } V = \text{const}, \lambda = \text{constant}\right] \tag{57}$$

5 Oscillations & Waves (Unit 10)

5.1 Simple Harmonic Motion (SHM) Kinematics & Energy

Governing Differential Equation, Velocity & Acceleration in SHM

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \implies x(t) = A\sin(\omega t + \phi) \tag{58}$$

$$v(t) = \frac{dx}{dt} = \omega\sqrt{A^2 - x^2}, \qquad a(t) = \frac{dv}{dt} = -\omega^2 x \tag{59}$$

$$v_{\max} = \omega A \quad (\text{at mean } x = 0), \qquad a_{\max} = \omega^2 A \quad (\text{at extreme } x = \pm A) \tag{60}$$

Energy Spectrum of Particle Executing SHM:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2), \qquad U = \frac{1}{2}m\omega^2 x^2 \tag{61}$$

$$E_{\text{total}} = K + U = \frac{1}{2}m\omega^2 A^2 = \text{constant for undamped SHM} \tag{62}$$

$$\langle K\rangle_{\text{time}} = \langle U\rangle_{\text{time}} = \frac{1}{4}m\omega^2 A^2 = \frac{1}{2}E_{\text{total}}, \qquad \langle K\rangle_{\text{position}} = \frac{2}{3}E_{\text{total}}, \quad \langle U\rangle_{\text{position}} = \frac{1}{3}E_{\text{total}} \tag{63}$$

5.2 Standard Oscillating Systems & Combinations

Time Period of Springs, Pendulums & Liquid Columns

$$T_{\text{spring}} = 2\pi\sqrt{\frac{m}{k}}, \quad k_s = \frac{k_1 k_2}{k_1 + k_2}\ [\text{Series}], \quad k_p = k_1 + k_2\ [\text{Parallel}] \tag{64}$$

$$T_{\text{cut spring}} \implies k' = nk \implies T' = \frac{T}{\sqrt{n}} \quad [\text{Spring cut into } n \text{ equal pieces}] \tag{65}$$

$$T_{\text{simple pendulum}} = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}}, \qquad g_{\text{eff}} = |\mathbf{g} - \mathbf{a}| \quad [\text{Lift accelerating upward: } g_{\text{eff}} = g + a] \tag{66}$$

$$T_{\text{two block}} = 2\pi\sqrt{\frac{\mu}{k}}, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2} \quad [\text{Reduced Mass Formulation}] \tag{67}$$

$$T_{\text{U-tube liquid}} = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{L}{2g}} \quad [h = \text{equilibrium column height}, L = \text{total liquid length}] \tag{68}$$

5.3 Wave Mechanics, Speed of Sound & Organ Pipes

Wave Speed, Harmonics in Strings & Pipes

$$v = \frac{\omega}{k} = \nu\lambda, \qquad y(x,t) = A\sin(kx \mp \omega t + \phi) \quad [- \text{ for } +x \text{ direction}] \tag{69}$$

$$v_{\text{string}} = \sqrt{\frac{T}{\mu}}, \qquad v_{\text{gas}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \quad [\text{Newton-Laplace Correction}] \tag{70}$$

$$\nu_{\text{string},\,n} = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{T}{\mu}} \quad (n = 1, 2, 3, \dots) \quad [\text{All Harmonics Present}] \tag{71}$$

$$\nu_{\text{closed pipe},\,n} = (2n-1)\frac{v}{4L} \quad (n = 1, 2, 3, \dots) \implies \nu_1 : \nu_2 : \nu_3 = 1 : 3 : 5 \quad [\text{ODD Harmonics ONLY}] \tag{72}$$

$$\nu_{\text{open pipe},\,n} = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots) \implies \nu_1 : \nu_2 : \nu_3 = 1 : 2 : 3 \quad [\text{ALL Harmonics Present}] \tag{73}$$

Resonance Tube Experiment & Beat Frequency:

$$\lambda = 2(l_2 - l_1), \qquad e = \frac{l_2 - 3l_1}{2} \approx 0.6r, \qquad v = 2\nu(l_2 - l_1) \tag{74}$$

$$\nu_{\text{beat}} = |\nu_1 - \nu_2| \quad [\text{Waxing decreases frequency: } \nu\downarrow; \text{Filing increases frequency: } \nu\uparrow] \tag{75}$$

Wave Traps: Organ Pipes & Speed of Sound vs Pressure

- **Sound Speed Independence from Pressure:** $v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$. At constant temperature, doubling pressure doubles density, leaving the ratio P/ρ and sound speed v **completely unchanged**.
- **Closed Pipe Odd Harmonics:** In a closed organ pipe of length L, the 3rd harmonic is $\nu = 3\left(\frac{v}{4L}\right)$ which is the **1st overtone**. The n-th overtone is the $(2n+1)$-th harmonic!

End of Module 02 Formula Handbook

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